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Laboratory application of NaI(Tl) γ -ray spectrometry to studies of natural radioactivity in geophysics

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Abstract

We implemented a laboratory apparatus based on a NaI(Tl) γ -ray spectrometer for the assessment of naturally occurring radioactive materials. This apparatus is devoted to the quantitative determination of U, Th and K in rock and soil samples. We analysed the calibration procedures with particular emphasis to factors which affect accuracy and detection limits. A first series of measurements were compared to those obtained by means of high energy γ -ray spectrometry. A discussion was made on factors restricting the approach based on scintillation spectrometry, such as secular equilibrium and radon encapsulation. © 2000 Elsevier Science Ltd. All rights reserved.

Keywords: Scintillation spectrometry; Calibration procedure; Radon effect; Secular equilibrium

1. Introduction

The γ -ray spectrometry method for the determination of naturally occurring radioactive materials (NORM) is of great interest in Earth's sciences. There is a number of possible applications spanning from ore exploration to environmental radiation monitoring problems, most of them involving the determination of the U, Th and K amount in soils and rocks. These elements may be used as tracers also in non-radioactive processes producing paragenesis associated with NORMs (Subjender Reddy and Venkat Rao, 1995). In geophysical studies, the heat produced by radioactive decay in rocks is of fundamental importance in under-

standing the thermal history of the Earth and interpreting the continental heat-flux data. To this regard, ^{40}K , ^{235}U , ^{238}U and ^{232}Th are the most important radioactive nuclides. Such heat sources contribute significantly to the heat flowing out from the Earth, since they are abundant, their heat production is sufficiently large and they have been effective during a time comparable with the age of the Earth (Rybach, 1988; Verdoya et al., 1998).

Van Schmus (1995) recently gave a compilation of the abundance of NORMs in terrestrial materials. As one of the alkali metals (Li, Na, K, Rb, and Cs), with an ionic radius of about 1.6 Å, potassium is one of the so-called large ion lithophile (LIL) elements. The isotopic abundance of ^{40}K , in weight, is 0.01193%. Potassium tends to get concentrated in crystal rocks, such as granites and shales, and in general is a primary constituent of several common rock-forming minerals.

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Most of natural uranium is formed by ^{238}U (99.28%) and ^{235}U (0.71%). Natural thorium is, instead, totally constituted by ^{232}Th . Also uranium and thorium are LIL elements, but they are found only in accessory minerals in common rocks or as minerals in primarily or secondary in deposits. Therefore, their overall abundance is much lower.

The main component of magmatic rocks like feldspar, quartz, olivine hornblende, pyroxene and mica are practically free of uranium and thorium. The accessory minerals, such as zircon, rutile, titanite and others have greater atom radii in their original lattice, and uranium and thorium can be substituted for the original atoms. During the process of magma solidification the main portion of U and Th is concentrated in the remaining, more acid portion of the liquid magma, which finally solidifies as granite. The particular temperature–pressure history during the solidification is decisive for the element content (Chiozzi et al., 1998).

This paper deals with the recent implementation of a NaI(Tl) γ -ray spectrometer at the Geothermal Laboratory of the University of Genova for the assessment of NORMs of relevant interest in terrestrial heat-flux studies. The equipment was calibrated taking into account the possible interference of each nuclides. A series of quantitative determination of ^{40}K , ^{238}U and ^{232}Th in several types of rocks was then carried out. The results were compared with those obtained by means of high resolution γ -ray spectrometry.

2. Experimental

2.1. Measuring system

Our γ -ray scintillation spectrometry system is an EG & G-Ortec NaI(Tl) detector (7.62×7.62 cm), connected to a multichannel analyser PC board (2048 channels) equipped with a spectrum stabiliser for automatic gain shift compensation. The detector is surrounded by a 5 cm thick lead shield which smoothes the background γ -radiation. Polyethylene Marinelli beakers were used both for standards and analysed materials. A volume of 500 ml was adopted with sample mass ranging approximately from 0.650 to 0.850 kg.

2.2. Reference materials

The system calibration was done by using three reference materials, obtained from the International Atomic Energy Agency (IAEA, 1987) for K, U and Th activity measurements. The potassium calibration standard (RGK-1) is extra-pure potassium sulphate (99.8%), with 9869 Bq of ^{40}K and uranium and thorium content lower than 0.001 and 0.01 ppm, respectively. The uranium standard (RGU-1) is U-ore diluted

with silica with ^{238}U activity of 3527 Bq, a negligible amount of K (<0.00234 ppm) and some traces of Th (<1 ppm). The thorium standard (RGTh-1) is Th-ore diluted with silica, with ^{232}Th activity of 2298 Bq, but containing some ^{238}U (18 Bq) and ^{40}K (0.07 Bq). The combined standard uncertainties on the activity values are 0.7%, 0.5% and 2% for K, U and Th, respectively.

In addition to these standards, a reference material was used for a better energy calibration, particularly in the medium-low energy portion of the spectrum. This material consisted of natural sand, spiked with a mixture of gamma-emitters, prepared and certified by the Istituto Nazionale di Metrologia delle Radiazioni Ionizzanti (ENEA, Rome), the Italian national institute for ionising radiation quantities.

2.3. Calibration and measurements

Energy calibration was performed in the energy range from 0.5 to 3 MeV using natural (dolomite and riolyte) and artificial (ENEA reference material) radionuclide sources. The following photon emissions, observed as single or double peaks, were used: ^{22}Na and ^{65}Zn annihilation peak (511.0 keV), ^{137}Cs (661.7 keV), $^{60}\text{Co} + ^{65}\text{Zn}$ (1144 keV), ^{88}Y (1836.1 keV), ^{40}K (1460.8 keV), $^{208}\text{Tl} + ^{214}\text{Bi}$ (596 keV), $^{228}\text{Ac} + ^{214}\text{Bi}$ (967 keV), ^{214}Bi (609.3, 1120.3, 1764.5 keV), ^{214}Pb (786.0 keV), ^{208}Tl (583.2, 2614.5 keV).

The measurement procedure described by Rybach (1971, 1988) was adopted, with some modifications intended to generalise the calibration aspects. Three counting windows or regions of interest (ROI) in the γ -ray spectrum were identified. They were centred on the three characteristic photo-peaks, at approximately 1.46 MeV (^{40}K), 1.76 MeV (^{214}Bi) and 2.62 MeV (^{208}Tl), corresponding to the ^{40}K , ^{238}U and ^{232}Th decay series (Fig. 1). Rybach (1988) suggested that ROIs should be as wide as 10% of the maximum energy peak of a characteristic nuclide. However, we preferred a slightly wider interval (15%), so that the full peak area was taken into account. A constant counting time, both for calibration sources and for measuring samples, of 5400 s was adopted.

The procedure proposed by Rybach for efficiency calibration can be generalised. The net count rate $\alpha_{i,j}$ in the i th ROI of a calibration standard j (with i and j equal to 1, 2 and 3 denoting the ROIs, and the calibration standards of K, U and Th, respectively) is proportional to the activity $A_{n,j}$ of each investigated nuclide n ($n = 1, 2$ and 3 for ^{40}K , ^{238}U and ^{232}Th , respectively) according to

$$\alpha_{i,j} = \sum_{n=1}^3 e_{i,n} A_{n,j} \quad (1)$$

where $e_{i,n}$ is the counting efficiency in the i th ROI for the nuclide n . The net count rate is given by

$$\alpha_{i,j} = \frac{N_{i,j}}{t_j} - R_{i,b} \quad (2)$$

where t_j is the counting time for the standard j , $N_{i,j}$ is the corresponding number of counts and $R_{i,b}$ is the background count rate in the i th ROI.

Eq. (1) is a linear system of $(i = 3) \times (n = 3)$ simultaneous equations that can be solved with respect to the nine counting efficiencies, on the basis of the measured count rates and activity of the standards. Provided that the calibration standards have highly unbalanced K/U, Th/U and K/Th ratios, this system of equations can be univocally solved.

Eq. (1) can be also used to obtain the activity of ^{40}K (A_K), ^{238}U (A_U) and ^{232}Th (A_{Th}) of a rock sample from the net count rate measured in the three ROIs (α_K , α_U and α_{Th}) and the counting efficiencies. In common applications, as proposed by Rybach, it is customary to neglect the contribution of K in the U and Th ROIs, since the latter are practically unaffected by the ^{40}K γ -ray spectrum (Rybach, 1971, 1988). Thus, A_K , A_U and A_{Th} can be obtained by solving the system of simultaneous equations:

$$\alpha_K = e_{1,1}A_K + e_{1,2}A_U + e_{1,3}A_{Th}$$

$$\alpha_U = e_{2,2}A_U + e_{2,3}A_{Th}$$

$$\alpha_{Th} = e_{3,2}A_U + e_{3,3}A_{Th} \quad (3)$$

provided that only seven calibration constants ($e_{1,1}, \dots, e_{3,3}$) are known.

The characteristics of the standard spectra simplify the calculation of the calibration constants. The adopted standards RGU-1 and RGK-1 are almost pure U and K_2SO_4 , so that the Th and K contribution, in the U standard, and the Th and U activity, in the K standard, can be neglected. In the Th standard there is a negligible amount of K, whereas U cannot be ignored. From the K standard, one has:

$$e_{1,1} = \frac{\alpha_{1,1}}{A_{1,1}} \quad (4)$$

from the U standard:

$$e_{1,2} = \frac{\alpha_{1,2}}{A_{2,2}} \quad \text{and} \quad e_{2,2} = \frac{\alpha_{2,2}}{A_{2,2}} \quad (5)$$

and from the Th standard:

$$e_{1,3} = \frac{\alpha_{1,3} - \frac{\alpha_{1,2}}{A_{2,2}}A_{2,3}}{A_{3,3}} \quad \text{and} \quad (6)$$

$$e_{2,3} = \frac{\alpha_{2,3} - \frac{\alpha_{2,2}}{A_{2,3}}A_{2,3}}{A_{3,3}}$$

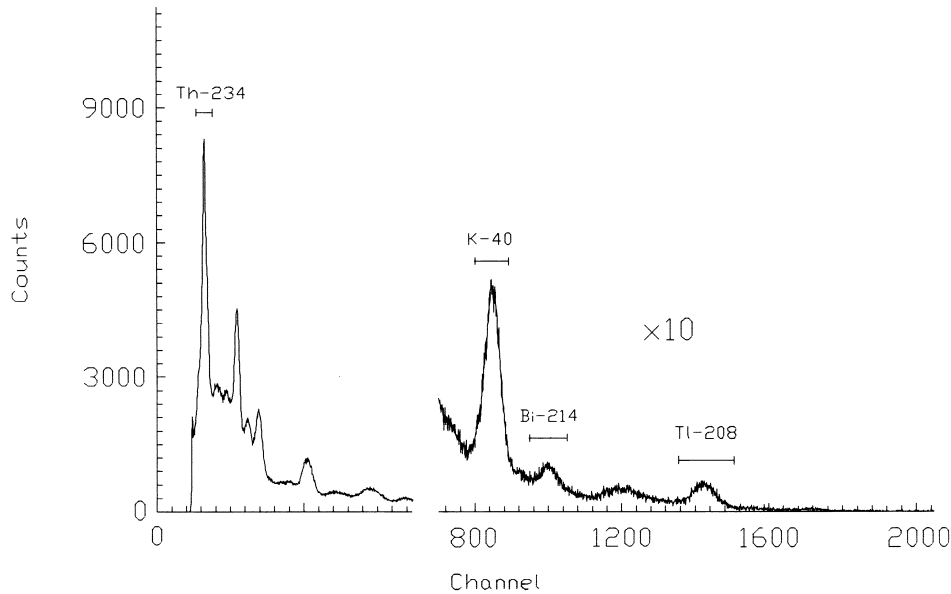


Fig. 1. γ -ray spectrum of sample R1. The energy windows utilised to determine K, U and Th concentration are indicated at the characteristic peaks. Counting time is 5400 s.

The other constants can be obtained by combining the count rates for the U and Th standards in the third ROI

$$e_{3,2} = \frac{\alpha_{3,2}}{A_{2,2}} \quad \text{and} \quad e_{3,3} = \frac{\alpha_{3,3} - \frac{\alpha_{3,2}}{A_{2,2}} A_{2,3}}{A_{3,3}} \quad (7)$$

A set of volcanic and sedimentary rocks were crushed, dried and sealed in polyethylene Marinelli beakers. They were counted and the spectra were analysed by means of the generalised method (Eq. (1)) and the simplified approach (Eq. (3)). Results are reported in Table 1. Analyses were based on the detection of γ -radiation emitted in the decay of ^{214}Bi (^{238}U series), ^{208}Tl (^{232}Th series), and ^{40}K . Common to the two approaches is that the determinations of U and Th were made under the assumption that the daughter nuclides are in equilibrium with the parent nuclides (Faure, 1986; Chiozzi et al., 1999). It resulted that the differences between the two approaches are smaller than 1/1000. This is a consequence of the simplification assumptions very close to the real characteristics of the calibration standards. Differently, particularly in field applications of this measurement technique where pure materials are difficult to obtain, the simplified method

is expected to give less reliable results (Chiozzi et al., this issue).

2.4. Measurement uncertainty and detection limits

Several factors may affect accuracy in the determination of the ^{40}K , ^{238}U and ^{232}Th activity. A correction for self-absorption due to the density variation in the analysed rock samples was attempted, but it resulted of negligible importance. Moreover, tests on the energy calibration revealed good stability (maximum drift of 6–8 keV). Therefore, the main sources of uncertainty were identified in counting statistics, calibration of the reference materials, mass determinations. Another critical point, the assumption of secular equilibrium in the ^{238}U decay series, will be discussed later.

The standard uncertainty on the activity values, σ_A , due to counting statistics is given by (Rybach, 1988):

$$\sigma_{A_K} = \frac{1}{e_{1,1}t} (N_1 + tR_{1,b})^{1/2}$$

$$\sigma_{A_U} = \frac{1}{e_{2,2}t} (N_2 + tR_{2,b})^{1/2}$$

Table 1

Activity concentration of uranium, thorium and potassium (in Bq/g) obtained by means of different measurement techniques: NaI(Tl) and HPGe γ -ray spectrometry^a

Sample	NaI(Tl)			HPGe		
	^{238}U	^{232}Th	^{40}K	^{238}U	^{232}Th	^{40}K
Basaltic andesite						
B1	0.027 (0.004)	0.035 (0.002)	0.857 (0.011)			
B2	0.024 (0.004)	0.020 (0.002)	0.502 (0.010)			
B3	0.023 (0.004)	0.027 (0.002)	0.507 (0.010)			
Andesite						
A1	0.046 (0.003)	0.048 (0.002)	0.984 (0.0119)			
A2	0.049 (0.003)	0.057 (0.002)	0.946 (0.011)	0.065 (0.005)	0.052 (0.002)	0.847 (0.028)
A3	0.042 (0.004)	0.051 (0.002)	0.899 (0.012)			
A4	0.041 (0.004)	0.054 (0.002)	0.724 (0.012)			
Rhyolite						
R1	0.130 (0.001)	0.147 (0.002)	1.292 (0.011)	0.194 (0.012)	0.142 (0.002)	1.305 (0.043)
R2	0.151 (0.002)	0.181 (0.003)	1.325 (0.015)			
Obsidian						
O1	0.183 (0.002)	0.219 (0.004)	1.379 (0.017)			
O2	0.167 (0.002)	0.193 (0.003)	1.368 (0.015)			
O3	0.175 (0.002)	0.197 (0.004)	1.314 (0.016)			
Dolomite						
	0.050 (0.001)	0.002 (0.001)	0.022 (0.003)	0.069 (0.003)	0.004 (0.001)	—
Granodiorite						
	0.046 (0.002)	0.056 (0.001)	0.881 (0.009)	0.064 (0.006)	0.050 (0.002)	1.016 (0.003)

^a HPGe data are from the National Institute of Metrology of Ionising Radiation (ENEA, Rome). In parenthesis the combined standard uncertainty of each value.

$$\sigma_{A_{Th}} = \frac{1}{e_{3,3}t} (N_3 + tR_{3,b})^{1/2} \quad (8)$$

where the subscripts 1, 2 and 3 of N and R are referred to the three investigated ROIs and t the counting time. The corresponding relative standard uncertainty components, for the investigated rocks of Table 1, is minimum for potassium (1.2%), maximum for uranium (3.5%) and amounts to 2.4% for thorium. This was confirmed experimentally. A set of 10 γ -ray spectra was acquired for the sample A2 (andesitic lava). The relative experimental standard deviation of the potassium, uranium and thorium specific activities resulted 1.8%, 3.9% and 3.1%, respectively.

The detection limit L_n of each nuclide n in a given sample depends on the background count rate, the acquisition time for that sample and the efficiency (Rybach, 1988):

$$L_n = \frac{2\sqrt{2}}{e_{i,n}} \left(\frac{R_{i,b}}{t} \right)^{1/2} \quad (9)$$

where $e_{i,n}$ and $R_{i,b}$ are defined as in Eq. (8) and t is the measuring time. A crucial point is the evaluation of the background radiation. At first we used a Marinelli beaker filled with ethylene glycol (density $\rho = 1110 \text{ kg m}^{-3}$). Finally, we adopted a powdered serpentinite with $\rho = 1710 \text{ kg m}^{-3}$, i.e. comparable to most of the investigated materials. This kind of rock may be virtually considered free of uranium and thorium and with a negligible amount (less than 0.03%) of potassium, thus generating a background count rate in the three ROIs more reliable than that obtained with the ethylene glycol (Chiozzi et al., this issue). By adopting the serpentinite spectrum as the background, the detection limits, for measurement times of one day and a sample of 0.750 kg, are 3.0, 0.7 and 0.7 Bq for ^{40}K , ^{238}U and ^{232}Th , respectively.

Table 2
Relative total count rate variation (%) versus the time elapsed from the initial value (just after sealing the sample holder) for some typical volcanic rocks^a

Sample	$\alpha_0 \text{ (s}^{-1}\text{)}$	Time (day)					
		2	6	9	14	16	20
Basaltic-andesite							
B1	56.84	1.14	1.09	2.01	0.94	1.29	1.38
Andesite							
A1	71.12	0.43	0.67	0.75	0.18	1.06	1.29
A2	72.74	0.54	0.32	0.99	0.67	0.24	0.57
A4	58.59	1.26	1.75	2.92	1.01	1.41	2.30
Rhyolite							
R1	156.28	0.40	0.72	1.28	0.72	0.63	0.90

^a α_0 is the initial count rate.

3. Discussion

Secular equilibrium between ^{226}Ra and ^{222}Rn (^{238}U series) can be practically assumed after about five to seven times the half-life (3.82 d) of the daughter ^{222}Rn . Aware of the fact that polyethylene is permeable to radon, we monitored the total spectrum count rate versus time for five volcanic rocks (basaltic-andesite, andesite and rhyolite). The first count was performed just after sealing the Marinelli beaker and the different samples were measured at the same days. The relative total count rate variation with time for a period of 20 days is reported in Table 2. There is a general increase of the total count rate until the ninth day up to a maximum deviation (for sample A4) of 2.92% from the initial value. Successively, the total count rate decreases in all the analysed samples. After 20 days, it increases again up to a maximum deviation of 2.30% (again for sample A4). At the moment, being the sources of instability of the counting system (counting efficiency, gain drifts) under experimental control, ^{222}Rn diffusion across the Marinelli beaker seems to be the most probable explanation of the observed systematic fluctuations. The use of metal sample holders could reduce this problem, but it does not provide a definitive solution for lack of secular equilibrium in the analysed samples as this may occur in other points of the decay chain.

The reliability of our NaI(Tl) detector was tested by a comparison with high resolution γ -ray spectrometry measurements, performed with an HPGe detector, for four selected rock samples in the same Marinelli beakers. The HPGe activity measurements were made for ^{238}U , ^{232}Th and ^{40}K . For ^{238}U , only the photon emissions of ^{234}Th and $^{234\text{m}}\text{Pa}$ were used, in order to not be involved with ^{222}Rn secular equilibrium problems. For ^{232}Th activity measurements, the photon emission of ^{228}Ac , ^{212}Pb , ^{212}Bi and ^{208}Tl were used and the single results, assumed as a measure of the ^{232}Th activity, were averaged. The single photon emission at 1460 keV was used as a direct measure of the ^{40}K activity.

The result of this comparison is reported in Table 1. No systematic deviation can be observed in the activity values of the two techniques. On average, the values obtained by HPGe spectrometry are higher by only 1% with a maximum deviation of 25%. Excluding the dolomite sample, wherein Th is close to the detection limit, thorium is slightly lower in HPGe determinations, the average difference being 8%. The maximum relative difference occurs for uranium (25–33%). It should be stressed that some of the investigated rocks (volcanics) are younger than 0.3 million years, and therefore they are suspected of ^{226}Ra disequilibria (Chiozzi et al., 1998). This result could indicate some secular disequilibrium in the decay series of ^{238}U (not affecting the HPGe ^{238}U measurement that were based

on its short half-life daughters). In the granodiorite and dolomite samples (age > 225 million years) weathering and leaching could have played a major role causing a ^{238}U apparent decrease in NaI(Tl) determinations by 28%. Moreover, as already pointed out, radon escape from the beaker might also affect the NaI(Tl) measurements.

Subjender Reddy and Venkat Rao (1995) and Ketcham (1996) suggested the use of the low energy part of the γ -ray spectra (0.03–0.10 MeV) for uranium determination, where there are a number of γ -rays produced by ^{234}Th (Fig. 1). In most cases it can be safely assumed that ^{234}Th in secular equilibrium with ^{238}U . This also gives lower detection limits. On the basis of these considerations, we attempted an efficiency calibration using the ^{234}Th ROI. The preliminary results indicated that this could smooth the relative difference between HPGe and NaI(Tl) spectrometry, since the difference in the specific activity of uranium obtained by means of the scintillator is on average lowered to about 10%.

4. Conclusions

Although the use of generalised calibration procedure for NaI(Tl) γ -ray spectrometry is in principle preferable, our results show that a simplified approach can be adopted, provided that almost pure K, U and Th standards are available. The equipment implement in our laboratory offers good detection limits and acceptable uncertainty in activity determinations allowing its use in geophysical applications such as the determinations of concentration of heat production elements in rocks and soils.

A comparison with a high resolution γ -ray spectrometry measurements confirms the reliability of our apparatus with some problems for the determinations of uranium. The differences could be due both to radon escape from the beakers and to weathering processes. In some cases, they could be ascribed to secular

radioactive disequilibrium due to the young age of the analysed samples. The use of the low energy portion of the γ -ray spectra could minimise the effects of the secular disequilibrium.

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